Arpan Pathak

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Visa: Requires H1B Transfer sponsorship

PERSONAL PROJECTS

Driving-CivicSense (Edge AI Vision) | github.com/arpanpathak/driving-civicsense-vision-model | 2026

- On-device AI perception system built in ***Rust*** for intersection discipline and lane awareness, running entirely on-device (***< 50ms inference latency***, no cloud dependency) for aftermarket dashcam and AR-glasses form factors.
- Fine-tunes ***YOLOv8/v11*** and custom ***CNN*** architectures; applies model pruning and quantization to fit tight memory and power budgets on low-powered single-board computers (Raspberry Pi 5).
- Trains on cloud GPU infrastructure with synthetic data augmentation; planning to release a public driving-behavior dataset on ***Hugging Face***.

Physics & Mathematics of Game Development in Rust (Book) | arpanpathak.github.io/bevy-physics-book | 2026

- Authored an unpublished book on the ***physics and mathematics of game development in Rust***, deep-diving the math and systems behind real-time simulation.

SUMMARY

Software Engineer with nearly 8 years of industry experience architecting performance-critical backend systems, distributed infrastructure, and real-time ***ML inference*** pipelines at ***Amazon***, ***Microsoft***, and ***Oracle***. Deep expertise in ***Rust***, ***C++***, ***Go***, and ***Python***: lock-free concurrency, memory safety, low-latency streaming (***Kafka***), and on-device inference. Proven track record shipping high-throughput, resource-efficient systems at global scale; focused on systems programming for consumer electronics, where performance, memory budgets, and real-time responsiveness are the product.

EXPERIENCE

Software Developer II Microsoft, **Redmond, WA** June 2025 – Present

- ***Rust Migration (Legacy C++/C#)***: Spearheaded migration of the Information Protection service from a legacy ***unsafe C++*** and garbage-collected ***C#*** codebase to ***Rust***, shifting memory safety to compile-time. Completely eliminated ***use-after-free***, ***buffer-overflow***, and ***null-safety*** bugs in the migrated path. Delivered measurable performance gains over C#, ***~40% lower P99 latency***, ***~2x throughput***, and ***~3x lower memory consumption per request***, with ***6-figure annual cost savings***. Significantly increased developer productivity and confidence writing memory-safe code for MSEC.
- ***AI-Powered Policy Engine***: Built an AI-driven engine that auto-generates Cilium network policies and scans for misconfigurations, reducing manual policy review cycles from **4 hours** to near-real-time with zero policy drift across sovereign regions.
- ***Zero-Trust Sovereign Cloud Networking***: Architected ***zero-trust sovereign cloud networking*** using ***eBPF and Cilium-based network policies*** to ensure ***data residency*** and ***SecNumCloud*** compliance standards. Built ***Kubernetes operators*** to auto-generate network policies with CI/CD pipelines.
- ***Developer Workflows***: Established CI/CD pipelines, containerized microservices, and automated validation frameworks ensuring reliable multi-region releases.

TechStack: ***Rust, C++, Kubernetes, eBPF***

Senior Software Engineer Oracle Cloud Infrastructure, **Seattle, WA** Oct 2024 – Mar 2025

- ***High-Performance Engine***: Architected low-latency internals for a proprietary database engine, optimizing ***lock-free data structures*** to reduce contention in high-concurrency read-write paths, improving transaction throughput by ***45,000 ops/s***.
- ***Infrastructure-as-Code SDK***: Engineered a Terraform provider (developer-facing SDK) for Oracle Autonomous Database, replacing legacy control plane infrastructure with projected ***$1.2M/month cost savings***.
- ***Security & Key Management***: Integrated OCI Security Vault across control plane services for secure secret management and automated key rotation protocols.

TechStack: ***Java, Go, Terraform Provider, Oracle Autonomous Database***

Software Development Engineer II Amazon, **Seattle, WA & Hyderabad, India** Mar 2021 – Oct 2024

- ***Real-Time ML Inference Engine***: Architected an ***ensemble model pipeline combining XGBoost and Deep Learning*** for real-time ads & deal ranking across FreeVee, Twitch, MiniTV, Prime Video, and Amazon Retail, processing millions of events at ***100ms inference latency*** while driving ***5% CTR increase***, ***30%*** higher deal impressions, and ***$5.4M monthly revenue growth***.
- ***Event-Driven Ads Systems***: Designed and delivered event-driven systems for the ads publisher platform, delivering ***programmatic guaranteed ad deals*** via real-time ***change data capture (CDC)*** with ***< 50ms latency*** for ad consumer-side platforms handling auction and programmatic buying inventory.
- ***Distributed Systems & SDK Design***: Designed a uniqueness constraint indexing solution and ***SDK*** for a multi-tenant NoSQL datastore using ***two-phase commit***, achieving ***serializable isolation*** at ***3K writes/second***, a developer-facing library used across multiple publisher teams.
- ***Conversational AI (BERT)***: Developed a ***BERT-based inference service*** handling **5,000 QPS** at peak, reducing support ticket triage SLA from **7 days** to **2 hours**.
- ***Big Data Platform & Data Lake***: Built big data platform, data warehouse, and data lake infrastructure for Amazon WorkEvents data handling ***1 petabyte*** of data and serving OLTP use cases. Designed a ***domain-specific language (DSL) execution engine*** to deploy datasets to ***AWS QuickSight*** for business intelligence, using ***AWS Glue***, ***Kotlin Spark API***, and ***AWS CDK (TypeScript)*** to orchestrate job infrastructure and data refresh strategies.

<> *TechStack*: ***Kotlin, Java, Python, TypeScript, AWS CDK, React.Js, DynamoDB, Redshift, AWS, PostgreSQL, PyTorch, XGBoost***

Software Development Engineer Razorpay, **Bengaluru, India** Aug 2020 – Feb 2021

- Built UPI Payment Gateway Aggregator using ***Golang*** and Protobuf, processing ***1M+ hourly transactions***.
- Maintained ***99.99% uptime*** with ***150ms P99 latency*** under **50,000 TPM** peak loads.

<> *TechStack*: ***Go, Redis, PostgreSQL, Protobuf, Google Cloud, AWS***

Software Engineer Mindfire Solutions, **Bhubaneswar, India** Aug 2018 – Feb 2020

- Developed full-stack workflow tools and an ***AR-based product search microservice***, serving 10,000+ daily queries.
- Built administrative dashboards reducing data retrieval latency from 5 min to 1 min.

<> *TechStack*: ***Java, SpringBoot, Angular, MySQL, MongoDB, AWS***

SKILLS

Languages: Rust, C++, Go, Python, Java, Kotlin, C, TypeScript, Shell Script

Systems Programming & Performance: Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, Performance Engineering, Low-Latency Systems

Networking: eBPF, Cilium, XDP, Kernel-Level Networking, TCP/IP, UDP, TLS, DNS, HTTP/3, Socket Programming, Packet Processing, Network Performance Optimization

Distributed Systems & Streaming: Apache Kafka, Apache Flink, Apache Spark, Distributed Systems (consensus, replication, sharding), Real-Time Data Pipelines, gRPC, REST API Design

ML Inference & AI: PyTorch, TensorFlow, XGBoost, Transformer Models (BERT), ONNX Runtime, On-Device/Edge Inference, Model Pruning & Quantization, Computer Vision (YOLOv8/v11), Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Sequential Models, Natural Language Processing (NLP), Deep Learning, Recommendation Systems, MLOps

Cloud & Infra: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS), Kubernetes (EKS, AKS), Docker, Terraform, Kubebuilder, Istio

Data & Storage: DynamoDB, PostgreSQL, Redis, MongoDB, Amazon Redshift, Amazon S3

Core CS: Data Structures & Algorithms, System Design, Operating Systems, Computer Architecture

EDUCATION

Bachelor of Technology in Computer Science & Engineering
RCC Institute of Information Technology, Kolkata, India | 2018